

ASCII MASTER FLOW – PANEL 1/12: Immune-object recognition map

PATHOGEN / PRODUCT -> contains or presents ANTIGEN

EPITOPE -> specific recognised part of antigen

B-CELL RECEPTOR / ANTIBODY -> binds epitope

T-CELL RECEPTOR -> recognises presented antigen fragment

IMMUNOGEN -> antigen capable of provoking an immune response

ASCII MASTER FLOW – PANEL 2/12: Active passive antiviral triad

VACCINE -> antigen/code -> endogenous response -> memory

mAb -> ready-made antibody -> direct binding -> temporary effect

ANTIVIRAL -> replication-step inhibition -> pharmacological action

ONSET / DURATION / PURPOSE -> compare, never merge

ASCII MASTER FLOW – PANEL 3/12: Adaptive immunity process rail

ANTIGEN PRESENTATION -> lymphocyte activation

B CELL -> plasma cell -> antibodies

B / T CELL -> memory populations

HELPER T CELL -> coordination; CYTOTOXIC T CELL -> cellular killing

RULE -> antibody titre is not the entire immune response

ASCII MASTER FLOW – PANEL 4/12: Vaccine platform taxonomy

WHOLE PATHOGEN -> live attenuated | inactivated

SELECTED MATERIAL -> subunit | toxoid | VLP | conjugate

GENETIC CODE -> viral vector | mRNA | DNA

COMPARE -> antigen form | replication | cell location | manufacture

DO NOT IMPORT -> product approval, efficacy or present availability

ASCII MASTER FLOW – PANEL 5/12: Genetic-platform cellular map

VIRAL VECTOR -> engineered carrier delivers antigen code

mRNA + LNP -> CYTOPLASM -> translation

DNA -> NUCLEUS -> transcription -> mRNA

ANTIGEN EXPRESSION -> B/T response -> memory

TRAP -> active immunity does not mean identical delivery

ASCII MASTER FLOW – PANEL 6/12: Monoclonal antibody production chain

B CELL + MYELOMA -> HYBRIDOMA CLONE
OR SEQUENCE DISCOVERY -> ANTIBODY ENGINEERING
RECOMBINANT MAMMALIAN CELL LINE -> EXPRESSION
HARVEST -> PURIFICATION -> FORMULATION -> FILL-FINISH
TARGET BINDING != PROVEN CLINICAL BENEFIT

ASCII MASTER FLOW – PANEL 7/12: Biopharma manufacturing quality system

MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

UPSTREAM -> culture / fermentation / bioreactor

DOWNSTREAM -> capture / purification / clearance

FILL-FINISH -> sterile product and packaging

RELEASE -> identity / purity / potency / sterility / consistency

ASCII MASTER FLOW – PANEL 8/12: Cold-chain integrity loop

VALIDATED STORAGE -> monitored transport -> delivery point

DATA LOGGER / INDICATOR -> deviation detection

EXCURSION -> quarantine / assessment / disposition

CONTINGENCY -> power, equipment, route and stock response

OUTCOME -> integrity enables access; it does not prove uptake

ASCII MASTER FLOW – PANEL 9/12: Clinical and regulatory ladder

PRECLINICAL -> laboratory and animal evidence

PHASE I -> initial safety / dose

PHASE II -> expanded dose, safety and response evidence

PHASE III -> larger-population benefit and safety

AUTHORISATION -> MANUFACTURE / RELEASE -> PHASE IV SURVEILLANCE

ASCII MASTER FLOW – PANEL 10/12: Indian institution router

DBT -> biotech policy, missions and research support

BIRAC -> translation, enterprise and mission implementation

ICMR -> biomedical research and evidence

CDSCO / DCGI -> trials and new-drug / biologic regulation

IPC / PvPI -> pharmacovigilance coordination and safety learning

MUST REMEMBER: Vaccines train active immunity through antigen delivery, whereas...

ASCII MASTER FLOW – PANEL 11/12: Biosimilar comparability pyramid

FOUNDATION -> extensive analytical and quality comparability

NEXT -> functional and appropriate non-clinical comparison

NEXT -> justified clinical PK/PD, efficacy, safety, immunogenicity

AFTER USE -> pharmacovigilance and traceability

RULE -> BIOSIMILAR != GENERIC != AUTOMATIC INTERCHANGEABILITY

CLOSE DISTINCTION: Antigen is not antibody, active immunity is not passive immunity,...

ASCII MASTER FLOW – PANEL 12/12: Biopharma claim and answer firewall

DEFINE -> immune mechanism and product category

CLASSIFY -> platform / mAb / biosimilar

TRACE -> trial + regulation + manufacturing + cold chain

MAP -> CDSCO / DBT / BIRAC / ICMR / PvPI

QUALIFY -> efficacy != effectiveness != approval != access != impact

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform or molecule, target,...